```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

np.random.seed(42)

n = 2503

shopping = pd.DataFrame({
    "Purchase_Amount": np.random.randint(200, 5000, n),
    "Discount": np.random.randint(0, 50, n),
    "Page": np.random.randint(1, 20, n),
    "Gender": np.random.choice(["Male", "Female"], n),
    "Category": np.random.choice(
        ["Electronics", "Clothing", "Home"], n
    ),
    "Payment": np.random.choice(
        ["UPI", "Card", "Cash"], n
    ),
    "Delivery": np.random.choice(
        ["Standard", "Express"], n
    )
})

shopping["Final_Price"] = (
    shopping["Purchase_Amount"]
    - shopping["Purchase_Amount"] * shopping["Discount"] / 100
)

print(shopping.head())
print(shopping.info())
print(shopping.describe())

sns.jointplot(
    x="Discount",
    y="Final_Price",
    data=shopping,
    kind="hex"
)
plt.show()

x = "Page"
y = "Purchase_Amount"

sns.jointplot(
    x=x,
    y=y,
    data=shopping,
    kind="hex"
```

```
)
plt.show()

plt.figure(figsize=(6, 4))

sns.heatmap(
    shopping.corr(numeric_only=True),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")
plt.show()
```

```
   Purchase_Amount  Discount  Page  Gender  Category Payment  Delivery
\
0             1060        27     1    Male  Clothing     UPI   Express

1             3972        28     6  Female  Clothing     UPI  Standard

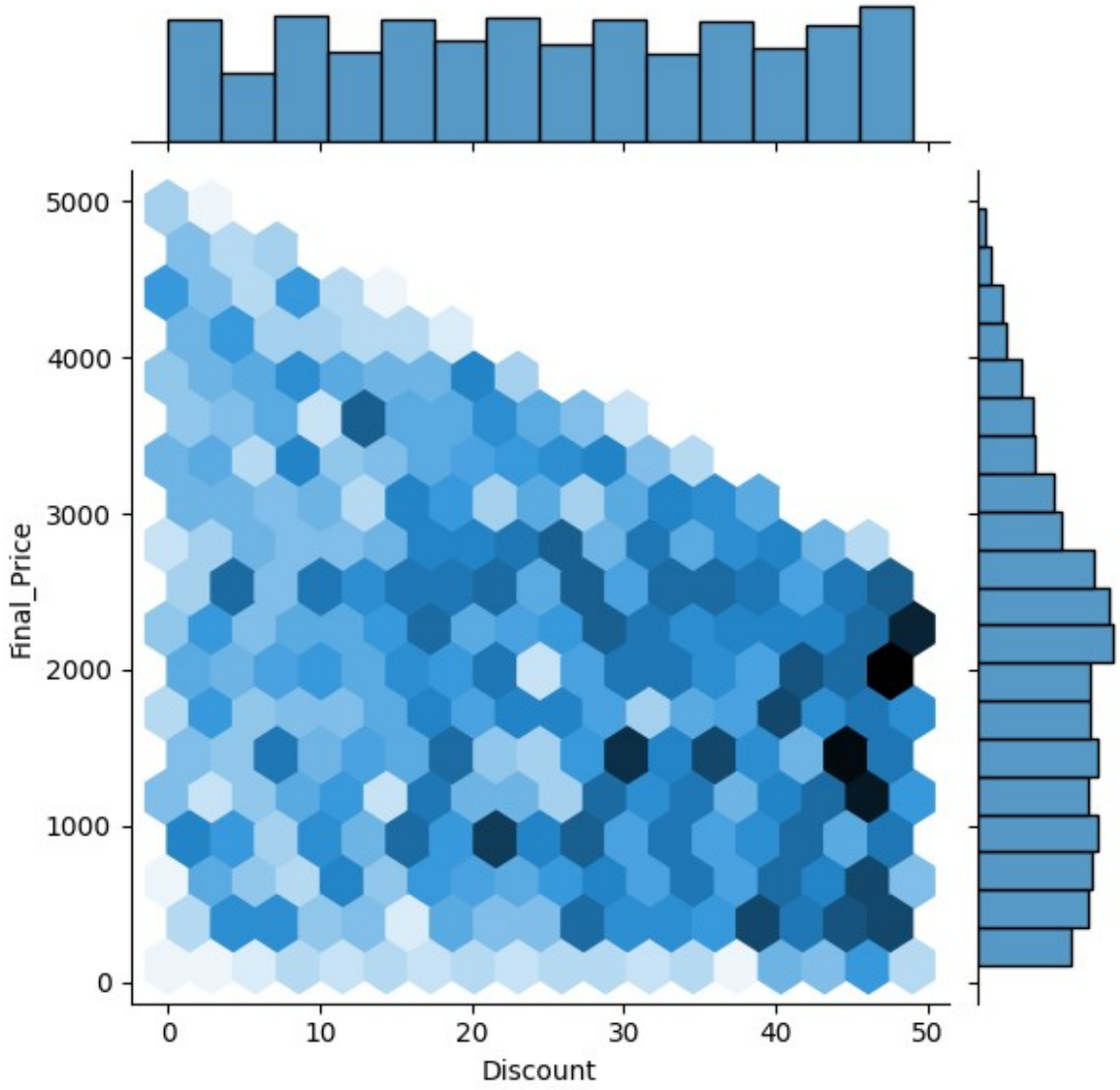
2             3292         8    15  Female  Clothing    Card   Express

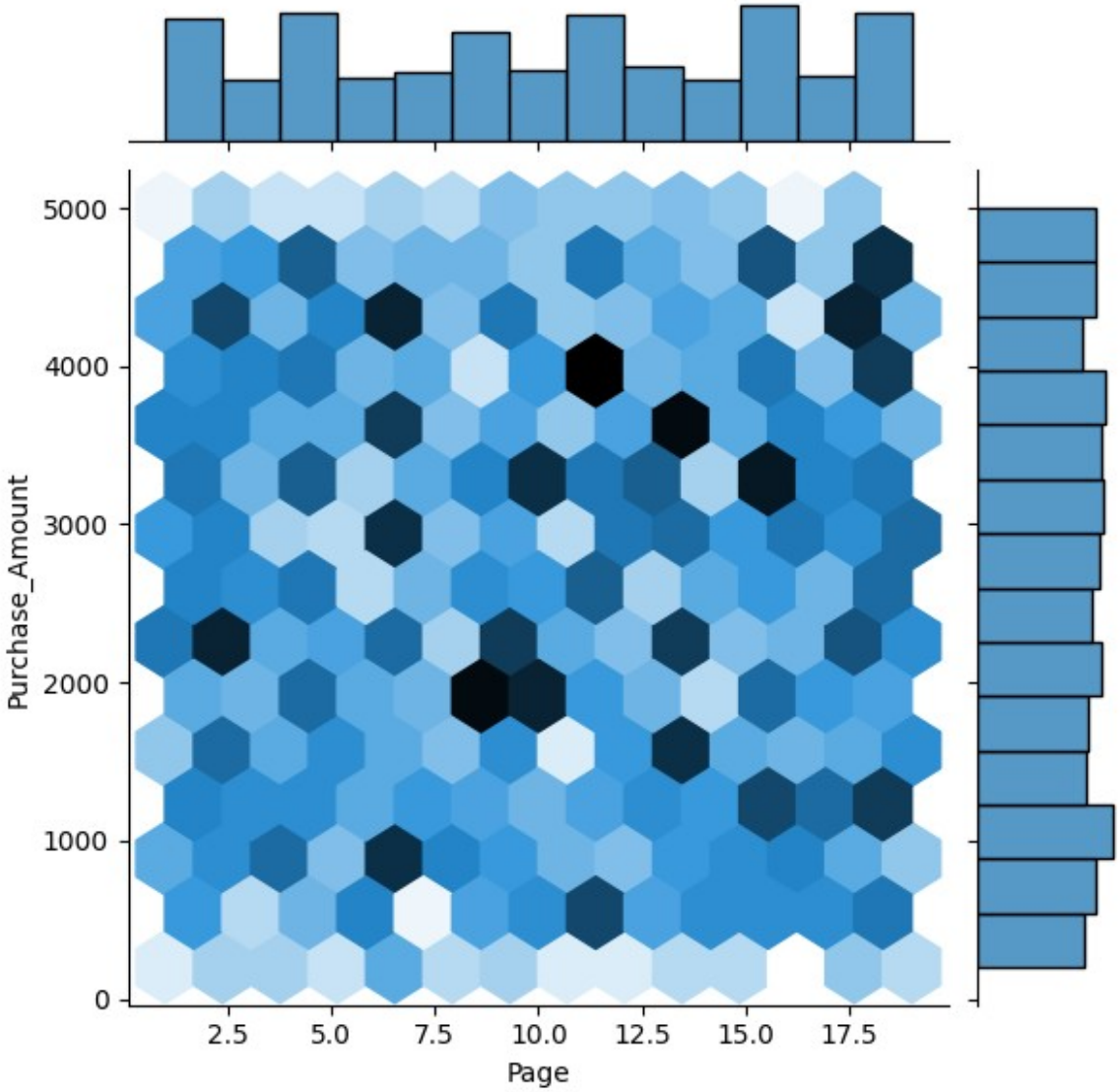
3              666        37    14    Male  Clothing    Card  Standard

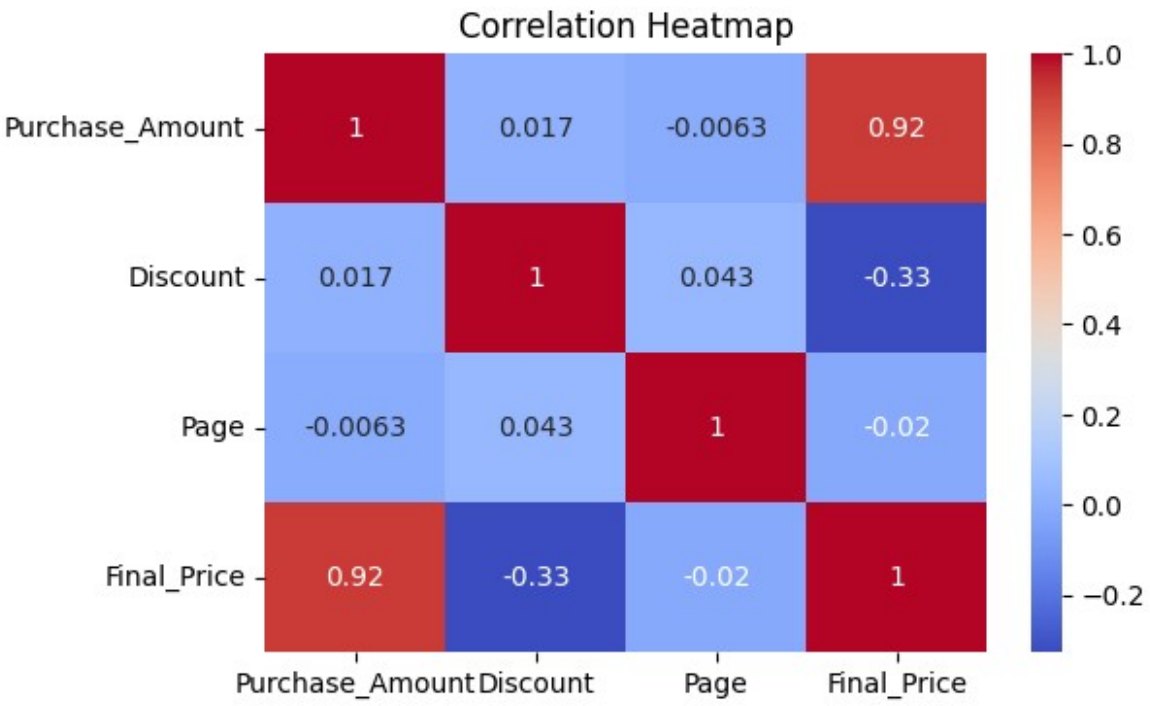
4             4626        49    15  Female      Home    Card   Express


   Final_Price
0       773.80
1      2859.84
2      3028.64
3       419.58
4      2359.26
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2503 entries, 0 to 2502
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Purchase_Amount  2503 non-null   int64
 1   Discount         2503 non-null   int64
 2   Page             2503 non-null   int64
 3   Gender           2503 non-null   object
 4   Category         2503 non-null   object
 5   Payment          2503 non-null   object
 6   Delivery         2503 non-null   object
 7   Final_Price      2503 non-null   float64
dtypes: float64(1), int64(3), object(4)
memory usage: 156.6+ KB
None
       Purchase_Amount     Discount         Page  Final_Price
```

count	2503.000000	2503.000000	2503.000000	2503.000000
mean	2612.968038	24.866560	10.136636	1959.905513
std	1370.582778	14.362769	5.462366	1105.941082
min	201.000000	0.000000	1.000000	109.080000
25%	1443.000000	12.000000	5.000000	1031.490000
50%	2633.000000	25.000000	10.000000	1921.830000
75%	3778.000000	37.000000	15.000000	2714.135000
max	4999.000000	49.000000	19.000000	4952.000000




5000
4000
3000
2000
1000
0
Purchase_Amount
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Page


Correlation Heatmap
Purchase_Amount
Discount
Page
Final_Price
1
0.017
-0.0063
0.92
0.017
1
0.043
-0.33
-0.0063
0.043
1
-0.02
0.92
-0.33
-0.02
1
Purchase_Amount
Discount
Page
Final_Price
1.0
0.8
0.6
0.4
0.2
0.0
−0.2